Kol's evolution equations tested episode by episode and the structure of the absorptivity

Willian Wistuba Melo da Cunha
Self-taught physics enthusiast

We test the ingredients of Kol's statistical evolution equations for the planar three-body problem with zero total angular momentum, dividing 24,000 free falls into 110,117 chaotic episodes separated by excursions. The excursions that return are reabsorbed by chaos with the same absorptivity measured by external launch, from 0.06 to 0.23 depending on the channel. The exits from chaos have almost no memory, and the escape probabilities with unequal masses are predicted within the statistical error, with a mean error of 0.6 to 0.9 percentage point, when the absorptivity and the measurement use the same definition of chaotic episode; the residual of the previous note came from inconsistent definitions. The dependence of the outcomes on the lifetime comes from the first episode and from the regular escapes, two channels predicted by the theory. The duration of the chaotic episodes, however, is not exponential. The absorptivity does not follow a simple law of the kick and the encounter speed, which explain at most 71% of its variation and fail for masses not used in the fit. Maps of the first passage show smooth regular bands, excursion zones that are also smooth and concentrate most of the absorptivity, and a small direct chaotic set with a fractal boundary. The absorptivity of a whole run equals the probability that the first passage ends in excursion or in chaos (correlation of 0.98 in 147 situations), which reduces the problem to a single encounter; in it, the choice between flyby and exchange depends almost only on the masses, but there is not yet a universal law.

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I. INTRODUCTION

Kol's flux theory has two parts [1]. The first states with what distribution the system leaves the chaotic region: the flux of phase volume multiplied by the absorptivity, the probability that the same state, traversed backward in time, enters chaotic motion. The second is a system of time-evolution equations that links three compartments: the chaotic region, the excursions (the third body moves away while bound to the pair and then returns), and the final states. In this model, the chaotic region has no memory and loses probability at a constant rate; each excursion returns after a time given by the outer orbit; and, on return, the state is absorbed again by chaos with probability equal to the absorptivity or follows a regular map to another exit state.

The first part has already been confirmed in three dimensions [2, 3] and, in note FR-TC-09, in the plane with zero total angular momentum, with measured absorptivity [9]. That note left a residual of about one percentage point in the ejection probabilities with unequal masses, and the outcomes showed a weak but significant dependence on the lifetime of the system. The second part of the theory, as far as we could find, has been compared with simulations only through the shape of the lifetime distribution [2, 4]. This note tests the ingredients of the evolution equations episode by episode, checks whether they explain the residual and the dependence on the lifetime, and, finally, examines whether the absorptivity, the only piece of the theory that still has to be measured by simulation, follows a simple law.

II. METHOD

A. Episodes

We use the regularized integrator of the previous notes [8], in planar motion with $G = 1$, zero total angular momentum, and start from rest, with the same random initial conditions as note FR-TC-09, for the mass groups (1, 1, 1), (1, 1, 3), (3, 3, 1), and (1, 2, 3), with 6,000 runs per group. Adapting the criterion of Manwadkar, Trani, and Kol [3], we call an excursion an interval in which the most bound pair has the third body bound to it (negative relative energy) and beyond the tidal radius $2a\,(2m_B m_3/m_1 m_2)^{1/3}$ for more than one period of the pair. This divides each run into episodes: chaotic phases separated by excursions. An episode is chaotic when it has two or more scrambles, the phases without any hierarchical pair of note FR-TC-09 [9]; with one scramble or none, it is a regular passage. The 24,000 runs gave 110,117 episodes. For each one we record the duration, the number of scrambles, and how it ends: in an excursion of which body and with what energy, or in the final escape.

B. Absorptivity per episode

The absorptivity was measured as in note FR-TC-09, launching the third body against ready-made pairs, on a grid of 11 values of the relative energy of the pair $x = E_B/E_0$, between 1.01 and 20, and 12 values of the eccentricity e, with 200 runs per cell in each of the seven channels with unequal masses (184,800 runs). To be consistent with the definition of epi-

sode, a launch counts as absorbed when it makes two or more scrambles before the first excursion; we also tested three or more.

C. Search for a law and arrival map

For each cell of each channel, we computed four quantities of the encounter: the distance of closest approach of the incoming orbit; the kick, that is, the velocity change that the third body imparts to the pair when passing at a distance of the order of the semimajor axis, $Gm_s/(d\,v_p)$, divided by the orbital velocity of the pair; the encounter speed, that is, the ratio between the passage velocity v_p and the orbital velocity, which measures whether the encounter is fast or slow compared with the revolution of the pair [6]; and the mass ratio. We fitted the absorptivity with each combination and tested universality by leaving one channel out of the fit and predicting it. Finally, for four types of arrival, we swept the orbital phase of the pair and the orientation of its periastron on a 100×100 grid (40,000 runs) and classified each point by what happens in the first passage.

III. RESULTS

A. Bodies returning from an excursion are absorbed as the theory requires

In the theory, the probability that a returning excursion is reabsorbed by chaos is the same absorptivity that governs the exits. For the excursions that return with pair energy just below the escape threshold ($0.95 < x < 1$), we compared the fraction that becomes a chaotic episode with the absorptivity measured on the escape side, just above the threshold ($x \leq 1.05$), weighted by the flux. These are two independent experiments: one observes returns within the free falls and the other launches bodies from outside against ready-made pairs. The values agree in all channels with sufficient sample (Table I). In the two channels with fewer than 20 returns, the uncertainty is too large for the test.

Most returns are not absorbed: 74% of the episodes after an excursion have a single scramble, that is, the body returns, passes by the pair, and leaves again without chaos. This is the regular map of the theory, and not a failure of it.

B. The exit from chaos has almost no memory

If the chaotic region has no memory, the chance that a chaotic episode ends in escape, rather than in excursion, must be the same in the first, the second, or the tenth episode. Table II shows that this holds for equal masses and for (1, 1, 3); in (1, 2, 3) and (3, 3, 1) there are deviations of a few percentage points, with modest significance. The duration of the previous excursion does not influence the exit, except in (3, 3, 1), where the correlation is weak ($\rho = 0.10$) but detectable. The mean duration of the chaotic episodes also does not change with the episode index.

One ingredient does not pass: the duration of the chaotic episodes does not follow the exponential distribution of a memoryless process. The Kolmogorov-Smirnov distance to the exponential is 0.19 to 0.26. With unequal masses, the coefficient of variation lies between 0.79 and 0.90, below 1, because an episode needs a minimum time to happen; with equal masses, it reaches 2.4, because some episodes contain long hierarchical phases that do not qualify as excursions. The excursions, in turn, last a median of 77% of the Kepler period of the outer orbit, with a slope of 0.83 on a logarithmic scale, consistent with the theory, given that we count only the stretch beyond the tidal radius.

C. With consistent definitions, the residual disappears

In note FR-TC-09, the absorptivity was measured by counting the scrambles of the whole run, including those that occur after excursions and returns, and the prediction was compared with all free falls with five or more scrambles [9]. The theory, however, predicts the distribution of exits from chaotic episodes, and the corresponding absorptivity is that of a single episode. Redoing the comparison consistently, that is, with the absorptivity counted up to the first excursion and with the escapes that leave a chaotic episode after the first, the residual disappears (Table III). The mean error over the five fractions falls to 0.6 to 0.9 percentage point, and all differences lie within 1.2 statistical errors.

D. Where the dependence on the lifetime comes from

In the evolution equations, the exit from a chaotic episode does not depend on the history, and therefore the lifetime can influence the outcome only through two paths: the first episode, which starts from an initial condition that has not yet been scrambled, and the escapes by regular map, in which a body returns from an excursion and leaves for good without new chaos. The data confirm this division. In the first episode, the outcome differs from that of the following episodes: in (3, 3, 1), a heavy body leaves in 22.3% of the first chaotic episodes and in 10.2% of the others; in (1, 2, 3), the light body leaves in 74.0% against 77.6%. In the escapes that leave a regular passage, the fraction of the heavy body grows with the lifetime in (1, 1, 3) and (3, 3, 1) ($p = 0.05$ and $p = 0.04$). In the escapes that leave a chaotic episode after the first, by contrast, there is no dependence on the lifetime in any of the three unequal groups (p from 0.30 to 0.48).

E. The absorptivity does not follow a simple law

Table IV summarizes the search for a law. The distance of closest approach explains almost nothing, because, with zero total angular momentum, almost all arrivals cross the pair. The kick alone explains little. The encounter speed explains about

Table I. Fraction of the excursions that return with $0.95 < x < 1$ and become a chaotic episode, with statistical error, compared with the absorptivity measured by external launch with $x \leq 1.05$, flux-weighted mean over e.

Masses	Returning body	Returns	Become chaos	Absorptivity
1, 1, 3	mass 1	703	0.132 ± 0.013	0.146
3, 3, 1	mass 1	720	0.219 ± 0.015	0.226
3, 3, 1	mass 3	60	0.067 ± 0.032	0.062
1, 2, 3	mass 1	733	0.181 ± 0.014	0.198
1, 2, 3	mass 2	95	0.116 ± 0.033	0.132

Table II. Fraction of the chaotic episodes that end in escape, by episode order from the second onward, and p-value of the chi-square test of independence.

Masses	2nd episode	3rd	4th and 5th	6th or later	p
1, 1, 1	0.116	0.108	0.098	0.098	0.28
1, 1, 3	0.226	0.232	0.253	0.239	0.50
1, 2, 3	0.258	0.230	0.219	0.206	0.009
3, 3, 1	0.272	0.279	0.303	0.231	0.012

half of the variation: encounters that are slow compared with the revolution of the pair almost never disrupt it, as expected for hard pairs [6]. Kick and encounter speed together reach 54% to 71%, while the product of the two, taken as a single number, explains 8% to 13%. The decisive test is universality, and it fails: when predicting a mass channel left out of the fit, the typical error is 85% of the spread of the absorptivity itself, even including the mass ratio.

F. The structure of the absorptivity

The phase and orientation maps (Fig. 1) show why no coarse combination suffices. Each map has three types of region. There are regular bands, smooth and wide, which occupy 51% to 87% of the map, in which the first passage has an immediate and certain result: the incoming body passes and goes away, or exchanges places with one of the members of the pair, which is expelled; in them there is no chaos, and only 1% to 2% of the points end in excursion (22% in the case of the light body arriving at a heavy pair). There are excursion zones, which occupy 13% to 43%, in which the first passage leaves a body bound but far away (94% to 99% of the points); the final outcome of these zones is decided later, and that is why they appear dotted. And there is direct chaos, which occupies only 4% to 6% (none in the case of the light body) and forms filaments and wedges at the edges of the excursion zones. The boundary of the chaotic set is fractal: the uncertainty exponent, which measures how the fraction of neighbors with different outcomes falls as the resolution is refined [7], is 0.34 to 0.51, which corresponds to a boundary of dimension 1.5 to 1.7. Direct chaos is not concentrated on the boundary between flyby and exchange.

In these four cases, the absorptivity of the whole run (0.22 to 0.47) is almost entirely the area of the excursion zones, while the absorptivity per episode (0 to 0.06) is the area of direct chaos. The first is large and has smooth edges; the second is small and fractal.

G. Everything is decided in the first passage

The maps suggest that the absorptivity of a whole run is already fixed in the first passage. To test this in all channels, we ran 20,160 arrivals (eight channels, including equal masses, seven values of x between 1.05 and 10, three values of e, and 120 runs per cell), integrated only up to the first exit, that is, up to the start of the first excursion or the detection of escape, with a limit of $30\,t_{\rm cr}$. In each cell we added the probability that the first passage is regular and leaves the outgoing body bound to the pair ($x_{\rm novo} < 1$, an excursion) and the probability that it is chaotic. In the 147 cells in which there is a measurement of the whole run, this sum reproduces the absorptivity of the whole run with a correlation of 0.979 and a mean deviation of 0.025 (means of 0.176 and 0.174); the chaotic portion reproduces the absorptivity per episode with a correlation of 0.91 and a mean deviation of 0.013. The absorptivity therefore reduces to what happens in a single passage, which costs about one third of the computing time of following the whole run.

Within each cell, the regular first passage is easy to describe. The logarithmic change in the energy of the pair, $D = \ln(x_{\rm novo}/x)$, follows approximately a Gaussian distribution, and the fraction below the threshold $D < -\ln x$ predicts the excursion probability with a mean error of 0.031, for a mean probability of 0.164. The center and width of this Gaussian, however, do not follow universal laws: fitted by the encounter speed or by the energy of the pair, by the eccentricity, and by the masses, they have R^2 of 0.52 to 0.59, but the prediction fails in the channels left out of the fit. There is, nevertheless, a clear separation (Table V). The choice between flyby and

Table III. Fraction of escapes per body among the escapes that leave a chaotic episode after the first (from 1,036 to 1,118 per group), with statistical error, compared with the flux multiplied by the absorptivity per episode (criteria of two and of three scrambles) and with the prediction of note FR-TC-09, which used the absorptivity of the whole run.

Masses	Ejected body	Measured	Predicted (per episode)	FR-TC-09 prediction
1, 1, 3	mass 3	0.026 ± 0.005	0.025	0.021
3, 3, 1	one of the masses 3	0.046 ± 0.006	0.053 to 0.054	0.073
1, 2, 3	mass 1	0.851 ± 0.011	0.861 to 0.868	0.833
1, 2, 3	mass 2	0.131 ± 0.010	0.115 to 0.121	0.148
1, 2, 3	mass 3	0.018 ± 0.004	0.017 to 0.018	0.019

Table IV. Fraction of the variation of the absorptivity between cells explained by each combination of encounter quantities (R^2), for the per-episode and whole-run criteria. The statistical noise limits the maximum possible to 0.94 and 0.99, respectively.

Quantities used	R^2
Distance of closest approach	0.02
Kick	0.11 to 0.20
Encounter speed	0.43 to 0.51
Kick and encounter speed	0.54 to 0.71
Kick × encounter speed (one number)	0.08 to 0.13
Energy and eccentricity of the pair	0.45 to 0.53

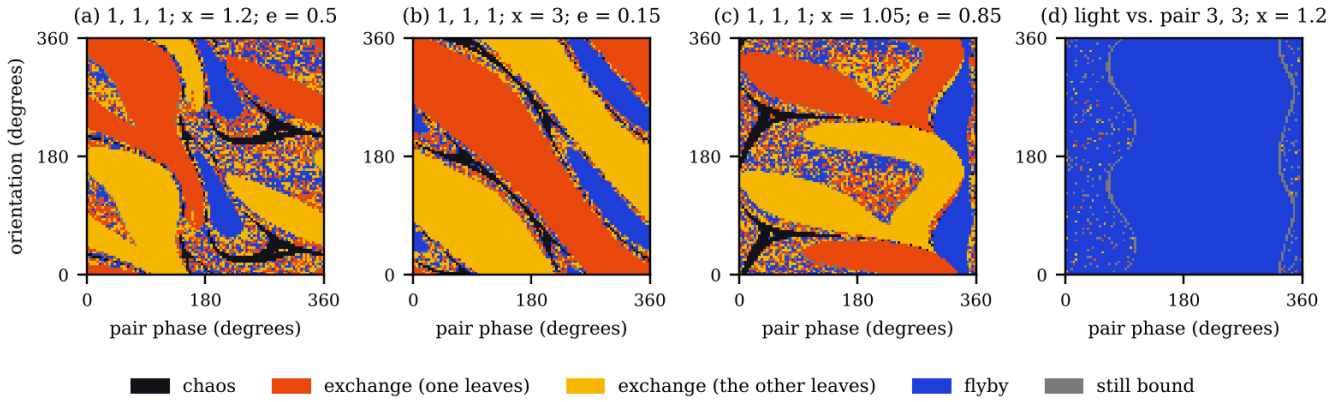


Fig. 1. Result of the first passage as a function of the orbital phase of the pair (horizontal) and the orientation of its periastron (vertical), for (a) equal masses with $x = 1.2$ and $e = 0.5$, (b) equal masses with $x = 3$ and $e = 0.15$, (c) equal masses with $x = 1.05$ and $e = 0.85$, and (d) the light body arriving at a pair of masses 3, with $x = 1.2$ and $e = 0.5$. Black: chaos in the first passage. Red and yellow: exchange, with expulsion of one or the other member of the pair. Blue: flyby. Gray: system still bound at the end of the integration. The dotted regions are excursion zones, whose final outcome is decided later.

exchange depends almost only on the masses, and not on the energy of the pair: the fraction of flybys varies little with x within each channel and goes from almost zero, when a heavy body arrives at a light pair, to 97%, when a light body arrives at a heavy pair. The probability that the outgoing body remains bound, on the other hand, falls with x in both branches, flyby and exchange. The rule that the lightest body is the one that leaves does not hold in the first passage: it is fulfilled in 36% to 100% of the cases, depending on the channel.

IV. DISCUSSION

Kol's evolution equations pass the central tests. The absorption of the returns is given by the same absorptivity that governs the exits, which confirms, across the boundary between escape and excursion, the reversibility relation on which the theory rests [1, 3]. The exits from chaos have almost no memory. And, with consistent definitions of chaotic episode on both sides of the comparison, flux times absorptivity predicts the escape probabilities with unequal masses within the statistical error. The residual of note FR-TC-09 came from

Table V. Fraction of flybys, versus exchanges, in the regular first passage, by channel: mean over the cells and range of variation across the values of x.

Pair	Incoming body	Fraction of flybys	Range in x
1, 1	1	0.17	0.11 to 0.22
1, 1	3	0.00	0.00 to 0.01
3, 1	3	0.05	0.03 to 0.06
1, 2	3	0.03	0.01 to 0.03
1, 3	2	0.32	0.30 to 0.35
1, 3	1	0.66	0.65 to 0.73
2, 3	1	0.86	0.82 to 0.90
3, 3	1	0.97	0.96 to 0.98

comparing quantities defined in different ways, and not from a failure of the theory. The dependence on the lifetime also fits the structure of the equations: it comes from the first episode, which retains memory of the initial condition, and from the escapes by regular map, two channels that the theory contains. What does not pass is the hypothesis of a constant decay rate at the level of each episode: the duration of the chaotic episodes is not exponential, although its mean does not depend on the history.

As for the absorptivity, the search for a simple law failed, but it left a clue. The encounter speed compared with the revolution of the pair is the quantity that weighs most, as in the adiabatic behavior of hard pairs [6], and the kick adds almost nothing. The maps show that the absorptivity of a whole run is almost entirely the area of the zones in which the first passage leaves a body bound. The test with all channels confirms that the absorptivity is fixed in the first passage, which reduces the problem of a whole chaotic history to a single encounter. This encounter does not yet have a formula, but it has structure: the choice between flyby and exchange depends on the masses, and the probability that the outgoing body remains bound depends on the energy of the pair. A formula, if one exists, must keep these two parts separate. Direct chaos, on the other hand, is small and fractal and should continue to require measurement.

The main limitations are as follows. The simulations are planar, with zero total angular momentum. The definitions of excursion, scramble, and chaotic episode are operational choices, and the quantitative results depend on them, even though the criteria of two and of three scrambles give similar results. With 6,000 runs per group, the sample of chaotic escapes after the first episode is about 1,100 per group, which limits the precision to about one percentage point. The phase maps cover only four types of arrival, and the first-passage test uses 120 runs per cell.

V. CONCLUSIONS

In the planar three-body problem with zero angular momentum, the ingredients of Kol's evolution equations pass the direct tests: the returns from excursions are reabsorbed with the same absorptivity as the exits, the exits from chaos have almost no memory, and the escape probabilities with unequal masses are predicted within the statistical error when the absorptivity and the measurement use the same definition of chaotic episode. The dependence of the outcomes on the lifetime comes from the first episode and from the regular escapes. The duration of the chaotic episodes is not exponential. The absorptivity does not reduce to a simple law of the encounter quantities, but most of it corresponds to smooth zones of the arrival space, in which the first passage leaves a body bound, and direct chaos occupies a small, fractal set. The absorptivity of a whole run equals the probability that the first passage ends in excursion or in chaos, which reduces its calculation to a single encounter; in this encounter, the choice between flyby and exchange depends almost only on the masses.

TRANSPARENCY AND USE OF ARTIFICIAL INTELLIGENCE

The author is a lawyer and not a physicist. This work results from independent research, conducted with the assistance of artificial intelligence (a language model) in the literature searches, the programming of the simulations, the analysis of the results, the writing, and the translation. The author formulated the questions, followed the experiments, and is responsible for the published content.

DATA AND CODE AVAILABILITY

All numbers were obtained with scripts published at https://fisica-revisitada.vercel.app/en/code/, which run on Node.js without external dependencies and allow the results to be reproduced.

CONFLICT OF INTEREST

The author declares no conflicts of interest. This work received no funding. This text has not undergone peer review and should be read as a working note. Corrections are recorded in the note itself, with the date.

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